

AIGR Gene Hypothesis Deep Research — CSR-1 (H2KZD5) "miRNA Binding" Function Assignment

Target gene: *csr-1* (CSR-1a / F20D12.1a), *Caenorhabditis elegans* (NCBITaxon:6239) **UniProt:** H2KZD5 (isoform A, 1030 aa); Q27GU1 (isoform B / CSR-1b, 867 aa) **Focus type:** function_assignment **Hypothesis slug:** corrected-argonaute-mirna-binding-specificity **GO term under evaluation:** GO:0035198 "miRNA binding" (MF)

Summary

The molecular-function annotation "CSR-1 binds miRNAs" (GO:0035198) on H2KZD5 is **not experimentally supported** for *C. elegans* CSR-1. It is a purely **phylogenetic (IBA) carry-over** annotation, propagated by GO_Central from a broad Argonaute-family ancestral node whose experimental anchors are the canonical bilaterian miRNA Argonautes — not CSR-1. Every primary experiment on CSR-1 tests **22G-RNA (endo-siRNA) binding and siRNA-directed slicing**, never miRNA loading. CSR-1's guide class (22G-RNAs) is produced by an RNA-dependent RNA polymerase (RdRP)/DRH-3 route that is **Dicer-independent**, mechanistically distinct from the Dicer→miRNA duplex pathway serviced by the worm miRNA Argonautes ALG-1/ALG-2.

The investigation confirmed the exact provenance of the annotation. QuickGO shows GO:0035198 on H2KZD5 carries evidence ECO:0000318 (IBA), reference GO_REF:0000033, and a `withFrom` field pointing to PANTHER internal-branch node **PTN001113179** — the precise node named in the seed hypothesis — anchored by *Drosophila* AGO1, mammalian AGO1–4, worm *alg-1/alg-2*, and zebrafish Argonaute. None of those anchors is *csr-1*. PANTHER v19 (family PTHR22891, subfamily SF181) is the propagation source that placed the miRNA-binding term onto CSR-1 as a family-level attribute.

The most important caveat, which the seed hypothesis correctly raises, is epistemic: no published assay has **directly tested and excluded physical miRNA binding** by CSR-1. The refutation is therefore of the *annotation's evidentiary basis* — there is no worm evidence for it,

and the evidence that exists is all about a different guide class — rather than a positive biochemical demonstration that CSR-1 cannot bind a miRNA duplex in vitro. For curation, the accurate statement is "no positive evidence for miRNA binding," which for a molecular-function term justifies **removal / non-core / NOT-qualification**. Separately, the accession/identity claims in the seed hypothesis are **correct**: H2KZD5 and Q27GU1 are the two genuine CSR-1 isoforms; Q17370 is *nhr-47* (carrying an obsolete "csr-1" synonym), and Q21992 is a deleted entry formerly belonging to LARP-1.

Executive Judgment

Verdict: Over-annotated / refuted as a direct CSR-1 function (with one honest residual caveat).

Where CSR-1 has real worm-based biochemical evidence, that evidence points to **siRNA (22G-RNA) activity**, not miRNA binding. The miRNA-binding term is the only one of CSR-1's molecular-function annotations that lacks an experimental (IDA/IMP/IPI) code — a strong signal that it is inference, not observation. The demonstrated, experimentally grounded molecular function is **GO:0070551 "endoribonuclease activity, cleaving siRNA-paired mRNA"** (IDA:WormBase), with biological process **GO:0090625 "siRNA-mediated gene silencing by mRNA destabilization"** (IDA). The recommendation is to treat GO:0035198 as **non-core/removable or NOT-qualified**, promote an informative **siRNA/22G-RNA binding** term, and retain the IDA-supported slicer activity.

Key Findings

Finding 1 — GO:0035198 on H2KZD5 is an IBA (phylogenetic) annotation, not an experimental one

UniProt H2KZD5 (*csr-1*, F20D12.1a, 1030 aa) carries the standard Argonaute architecture — a PAZ domain plus the nematode Piwi (PIWI) domain — and is placed by PANTHER in family **PTHR22891** (eIF2C/Argonaute family). Its GO:0035198 "miRNA binding" term is annotated with

evidence code **IBA:GO_Central** — *inferred from biological ancestor*, i.e. propagated from the phylogenetic reconstruction of the Argonaute family, which includes bilaterian miRNA-binding AGOs.

By direct contrast, the **experimentally supported** molecular function for CSR-1 is **GO:0070551 "endoribonuclease activity, cleaving siRNA-paired mRNA"** (evidence IDA:WormBase), and the experimentally supported biological process is **GO:0090625 "siRNA-mediated gene silencing by mRNA destabilization."** Where CSR-1 has real worm biochemical evidence, that evidence points to **siRNA (22G-RNA) activity**, not miRNA binding. The miRNA-binding term is the only one of these MF annotations that lacks an experimental code — a strong signal that it is inference, not observation.

Finding 2 — Every primary CSR-1 experiment recovers 22G-RNAs / siRNA-directed slicing, never miRNA binding

The two primary references named in the seed hypothesis, plus the most relevant modern isoform paper, all interrogate the siRNA branch:

- **PMID: 19804758** (Claycomb et al., 2009): CSR-1 immunopurification recovers small RNAs that are "**antisense to thousands of germline-expressed protein-coding genes**" — i.e. 22G-RNAs. This is a guide-class recovery experiment; it identifies what CSR-1 co-purifies with, and that is the 22G-RNA class, not miRNAs.
- **PMID: 18007599** (Aoki et al., 2007): in a *C. elegans* cell-free assay, "**An Argonaute protein, CSR-1, is responsible for the Slicer activity induced by secondary-type siRNAs.**" This tests slicing directed by **secondary siRNA guides**, not miRNA loading.
- **PMID: 34244496** (Nguyen & Phillips, 2021): both isoforms bind 22G-RNAs; CSR-1A associates preferentially with **spermatogenesis-specific 22G-RNAs** and CSR-1B with **oogenesis-specific 22G-RNAs**, with the CSR-1A-unique exon's arginine dimethylation controlling this **siRNA-binding specificity**.

Critically for the seed hypothesis's own framing: these are experiments about **recovered guide classes and relative slicing efficiencies**, not direct tests of miRNA binding. A strong 22G-RNA preference does not, by itself, logically exclude the physical possibility of miRNA binding. But it does mean there is **no positive worm evidence** for miRNA binding, and the entire experimental record for CSR-1 loads it with 22G-RNAs.

Finding 3 — CSR-1 identity is confirmed; *nhr-47* / LARP-1 accessions are not this Argonaute

UniProt lookups confirm the identity claims in the seed hypothesis exactly:

Accession	Gene / entry	Identity	Family	Is this the Argonaute?
H2KZD5	<i>csr-1</i> / F20D12.1a	CSR-1A, 1030 aa	PTHR22891 (Argonaute)	Yes (isoform A)
Q27GU1	<i>csr-1</i> / F20D12.1b	CSR-1B, 867 aa	PTHR22891 (Argonaute)	Yes (isoform B)
Q17370	NHR47_CAEEL (C24G6.4)	Nuclear hormone receptor <i>nhr-47</i> , carries obsolete "csr-1" synonym	PTHR24083	No (not an Argonaute)
Q21992	Inactive / deleted entry	Formerly LARP-1	—	No (deleted)

This matters for curation hygiene: any GO annotation that reached the *csr-1* review via Q17370 or Q21992 would be a **misattribution/database carry-over** and should be discarded. The two genuine CSR-1 Argonaute isoforms are H2KZD5 and Q27GU1.

Finding 4 — PANTHER (PTHR22891:SF181) directly propagates GO:0035198 to CSR-1 as a family-level term

The PANTHER v19 geneinfo record for H2KZD5 (family **PTHR22891**, subfamily **SF181**) lists in its molecular-function annotation set: GO:0003676 (nucleic acid binding), GO:0003723 (RNA binding), GO:0003727 (single-stranded RNA binding), **GO:0035198 (miRNA binding)**, and GO:0070551 (endoribonuclease activity cleaving siRNA-paired mRNA). PANTHER's protein-class assignment is PC00224 "translation initiation factor" (the historical eIF2C/Argonaute label), and Reactome carryover for the family includes "MicroRNA (miRNA) biogenesis." This is the mechanistic **source** of the IBA:GO_Central "miRNA binding" annotation seen in UniProt — the term rides on the family, not on CSR-1-specific data.

For comparison, the bona fide worm miRNA Argonautes **ALG-1** and **ALG-2** are experimentally defined: **PMID: 11461699** (Grishok et al., 2001) shows *alg-1* and *alg-2* (together with Dicer *dcr-1*) are required for the maturation and activity of the *lin-4* and *let-7* small temporal RNAs (miRNAs). CSR-1 has no comparable miRNA-function evidence.

Finding 5 — QuickGO pins the exact IBD node; its experimental anchors are miRNA-Argonautes, not csr-1

The QuickGO annotation API for H2KZD5 / GO:0035198 "miRNA binding" returns: evidence **ECO:0000318 (IBA)**, reference **GO_REF:0000033**, assignedBy **GO_Central**, with the **withFrom** field pointing to PANTHER node **PTN001113179** and its experimentally annotated members. Those members are the canonical miRNA Argonautes:

- FB:FBgn0262739 — *Drosophila* AGO1
- MGI:2446630 / MGI:2446632 — mouse Argonautes
- UniProtKB Q9UL18 / Q9UKV8 / Q9H9G7 / Q9HCK5 — human AGO1–AGO4
- WB:WBGene000000105 / WBGene000000106 — *C. elegans* **alg-1** and **alg-2**
- ZFIN — zebrafish Argonaute

This is **exactly the IBD node named in the seed hypothesis (PTN001113179)**. The worm experimental anchors driving the "miRNA binding" term across the family are **alg-1/alg-2**, the true miRNA Argonautes — **not csr-1**. CSR-1 receives the term only by virtue of sharing the ancestral node, not because any of its own members were experimentally shown to bind miRNAs. (The leaf node PTN000527417 named in the hypothesis was not separately dereferenced but is consistent with H2KZD5's PTHR22891:SF181 placement descending from PTN001113179.)

Mechanistic Model / Interpretation

The core of the evaluation is a distinction between two Argonaute sub-systems that both live in the *C. elegans* germline but use different guide classes generated by different biogenesis machinery:

miRNA branch (ALG-1/ALG-2)	22G-RNA branch (CSR-1)
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pri-miRNA (lin-4, let-7 ...)	target mRNA template
Drosha	RdRP (EGO-1) – Dicer-INDEPENDENT
pre-miRNA	unprimed 5'-ppp 22G-RNA synthesis
Dicer (DCR-1) <-- REQUIRED	DRH-3 helicase complex
~22 nt miRNA duplex (5'-monoP)	22 nt 5'-G antisense 22G-RNAs (5'-triP)
loaded into ALG-1/ALG-2	loaded into CSR-1 (GO:0070551 slicer)
translational repression / mRNA destabilization (developmental timing: lin-4, let-7)	chromosome segregation, holocentric kinetochore organization, germline transcript "tuning" (NOT silencing) [PMID 19804758, 27020753]

Key mechanistic separators:

1. **Biogenesis.** miRNAs are **Dicer-dependent** (DCR-1). 22G-RNAs are made by RdRP (EGO-1) with the DRH-3 helicase and are **Dicer-independent** (PMID: 19800275). A CSR-1 that loads RdRP-made 22G-RNAs is operating in a fundamentally different guide-supply pipeline than a miRNA Argonaute.
2. **Guide 5' chemistry and length.** 22G-RNAs are 22 nt with a 5' triphosphorylated guanosine; miRNAs are ~22 nt with a 5' monophosphate produced by Dicer. These are physically distinct substrates for the MID/PIWI loading pocket.
3. **Function.** CSR-1's demonstrated roles are chromosome segregation / holocentric kinetochore organization (PMID: 19804758), germline transcript **tuning** rather than silencing (PMID: 27020753), and meiotic homolog pairing (PMID: 37078421). None of these is a miRNA-type post-transcriptional repression of developmental-timing genes.

The "miRNA binding" GO term therefore represents a **phylogenetic leak**: a family-ancestral capacity, real in ALG-1/ALG-2 and in bilaterian AGOs, painted onto CSR-1 because it descends from node PTN001113179. The evidence that would be needed to keep it — direct physical loading of a miRNA duplex into CSR-1 — has never been generated in worm, and everything that *has* been generated points to the 22G-RNA branch.

GO decision table

GO term	Aspect	Current evidence on CSR-1	Recommended curator action
GO:0035198 miRNA binding	MF	IBA only (PTN001113179 carry-over); no worm experiment	Remove / mark non-core, or NOT-qualify. Not a demonstrated CSR-1 function
GO:0070551 endoribonuclease activity, cleaving siRNA-paired mRNA	MF	IDA:WormBase; slicer assay [PMID 18007599]	Retain — core molecular function
GO:0090625 siRNA-mediated gene silencing by mRNA destabilization	BP	IDA-supported; siRNA branch	Retain (note "tuning" nuance from [PMID 27020753])
siRNA binding (GO:0035197 / 22G-RNA binding)	MF	Strongly supported [PMID 19804758, 34244496]	Add / prefer as the informative binding term over "miRNA binding"

Evidence Matrix

Citation	Evidence type	Supports / refutes / qualifies	Claim tested	Key finding	Context
PMID: 19804758	Direct assay (IP-small RNA seq) + mutant phenotype	Refutes miRNA- binding as CSR-1's guide class	What small RNAs does CSR-1 bind?	CSR-1-associated 22G-RNAs are antisense to thousands of germline protein-coding genes; CSR-1 needed for chromosome segregation	<i>C. elegans</i> germline
PMID: 18007599	Direct assay (cell-free slicer)	Refutes / qualifies — activity is siRNA- directed	Which Argonaute slices under secondary siRNAs?	CSR-1 is responsible for Slicer activity induced by secondary-type siRNAs	<i>C. elegans</i> cell-free extract
PMID: 34244496	Direct assay (isoform IP)	Refutes / qualifies	Guide specificity of CSR-1 isoforms	CSR-1A ↔ spermatogenic 22G-RNAs, CSR-1B ↔ oogenic 22G- RNAs; arginine dimethylation sets siRNA specificity	<i>C. elegans</i> germline
PMID: 11461699	Mutant phenotype / genetic	Competing (identifies the real miRNA Argonautes)	Which worm AGOs handle miRNAs?	<i>alg-1/alg-2</i> + <i>dcr-1</i> required for <i>lin-4/let-7</i> stRNA maturation & activity	<i>C. elegans</i>
PMID: 19800275	Genetic / biogenesis		How are 22G- RNAs made?	22G-RNAs are RdRP/DRH-3- dependent,	<i>C. elegans</i> germline

Citation	Evidence type	Supports / refutes / qualifies	Claim tested	Key finding	Context
		Qualifies (mechanistic separator)		Dicer- independent	
QuickGO annotation API (H2KZD5, GO:0035198)	Database (evidence-code audit)	Refutes experimental basis	Is miRNA binding experimental?	Evidence = IBA (ECO:0000318), GO_REF:0000033, withFrom PANTHER PTN001113179, anchored by alg-1/alg-2 + bilaterian AGOs	Curation record
PANTHER v19 (PTHR22891:SF181)	Computational / evolutionary	Qualifies (source of term)	Where does the term come from?	miRNA-binding is a family-level MF term propagated to CSR-1	Phylogenetic
UniProt (H2KZD5, Q27GU1, Q17370, Q21992)	Database (identity)	Supports identity claims	Are the accessions correct?	H2KZD5/Q27GU1 = CSR-1; Q17370 = nhr-47; Q21992 = deleted LARP-1	Sequence records
PMID: 27020753	Mutant phenotype	Qualifies (function is tuning)	What does CSR-1 slicing do?	CSR-1 tunes (not silences) germline transcripts to ensure embryonic divisions	<i>C. elegans</i> germline/ embryo
PMID: 37078421	Mutant phenotype	Qualifies (function is chromosomal)	CSR-1 role in meiosis	CSR-1/CSR-2 + 22G-RNAs required for accurate homolog pairing	<i>C. elegans</i> meiosis

Citation	Evidence type	Supports / refutes / qualifies	Claim tested	Key finding	Context
				& unpaired silencing	
PMID: 25510497	Comparative / evolutionary	Qualifies	Conservation of CSR-1 pathway	CSR-1 22G-RNA pathway and nuclear role conserved across <i>Caenorhabditis</i>	<i>C. briggsae/ elegans</i>

Computed provenance artifacts (from executed UniProt / PANTHER v19 / QuickGO queries): `csr1_evidence_matrix.csv`, `csr1_go_decision_table.csv`.

Evidence Base — narrative

The **primary references the seed hypothesis itself names** are the strongest evidence against the miRNA-binding assignment, precisely because of what they *do not* test. Claycomb et al. ([PMID: 19804758](#)) is a guide-class recovery experiment: it defines the CSR-1-associated small-RNA population as 22G-RNAs antisense to germline genes. Aoki et al. ([PMID: 18007599](#)) is a slicer assay driven by secondary siRNAs. Neither presents a miRNA to CSR-1. The modern isoform study ([PMID: 34244496](#)) reinforces this by showing the *entire* binding-specificity machinery of CSR-1 (down to arginine dimethylation of the CSR-1A exon) is tuned for **22G-RNA** discrimination between spermatogenic and oogenic guides.

The **comparative reference** ([PMID: 11461699](#)) matters because it names the actual worm miRNA Argonautes: ALG-1 and ALG-2 mature and act on *lin-4/let-7*. This is the experimental footprint that the family term "miRNA binding" legitimately reflects — and it belongs to a different gene. The QuickGO evidence-code audit closes the loop by showing that the *withFrom* provenance of the CSR-1 term is exactly PTN001113179, anchored by *alg-1/alg-2* and bilaterian AGOs, not by any *csr-1* experiment.

The **supporting/qualifying references** ([PMID: 19800275](#), [PMID: 27020753](#), [PMID: 37078421](#), [PMID: 25510497](#)) establish that CSR-1's guide class is biogenetically distinct from miRNAs (RdRP/Dicer-independent) and that its downstream functions (transcript tuning, chromosome segregation, meiotic pairing, cross-species conservation) are not miRNA-type repression. Together they place CSR-1 firmly in the 22G-RNA endo-siRNA world.

Mechanistic Scope

The **immediate molecular function** at issue is *physical loading/binding of a small-RNA guide* — specifically whether CSR-1's MID/PIWI guide-binding pocket accepts a miRNA. This must be separated from three adjacent layers that the literature *does* document and that should not be conflated with miRNA binding:

- **Guide-class recovery** — IP-seq yields 22G-RNAs; a *composition* readout, not a binding-exclusion assay ([PMID: 19804758](#)).
- **miRNA-directed slicing** — CSR-1 slices, but assays show it does so under **siRNA/22G-RNA** guides, a *catalytic* readout ([PMID: 18007599](#), [PMID: 27020753](#)).
- **Physiological gene regulation** — germline transcript "tuning," chromosome segregation, meiotic pairing and fertility are pathway/organismal consequences, not the binding event.

No published assay isolates the miRNA-loading step for CSR-1. The seed hypothesis is right to insist on this separation: a strong 22G-RNA preference does not *logically* exclude miRNA binding, but the burden of proof lies with a positive binding assay that has never been done. The GO term "miRNA binding" is a molecular-function (binding) claim, and it currently has zero direct-assay support in worm.

Conflicts and Alternatives

1. **Paralog/family over-annotation (the primary alternative, and it wins).** The most parsimonious explanation for GO:0035198 on CSR-1 is family-level propagation from PTN001113179. The withFrom anchors are ALG-1/ALG-2 and bilaterian AGOs. This is textbook IBA carry-over, and it is the reason to down-weight the term. The nematode-specific Piwi signature and lineage-specific acquisition of RdRP-dependent 22G loading flag the divergence that makes the ancestral miRNA term inappropriate for CSR-1.

2. **The real worm miRNA Argonautes are ALG-1/ALG-2 (PMID: 11461699):** they, not CSR-1, are genetically required for lin-4/let-7 miRNA maturation and function. 22G-RNAs are RdRP/DRH-3-dependent and Dicer-independent (PMID: 19800275), a biogenesis route incompatible with the canonical Dicer→miRNA duplex-loading pathway.
 3. **Database carry-over via wrong accessions (ruled out but worth flagging).** Q17370 (*nhr-47*, carrying an obsolete "csr-1" synonym) and Q21992 (deleted, formerly LARP-1) are decoys — confirmed *not* the source of the H2KZD5 annotation, but curators should ensure no legacy annotation entered *csr-1* through them.
 4. **Isoform nuance.** CSR-1A (H2KZD5) and CSR-1B (Q27GU1) differ in the N-terminal exon and in guide-specificity control (PMID: 34244496), but both bind 22G-RNAs. There is no isoform for which miRNA binding is demonstrated; the term cannot be salvaged by assigning it to one isoform.
 5. **The honest residual (pro-hypothesis caveat).** No study has run a direct in vitro CSR-1 miRNA-binding assay and reported a negative result. So the term is *unsupported*, not *disproven at the bench*. This is why the recommendation is removal/non-core/NOT-qualification rather than a claim of proven incapacity.
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Limitations and Knowledge Gaps

Gap	What was checked	Why it matters for curation	What would resolve it
No direct in vitro miRNA-binding assay for CSR-1	Primary CSR-1 papers all use 22G-RNA/siRNA guides	Distinguishes "unsupported" from "disproven"; sets the correct qualifier (remove vs NOT)	EMSA / filter-binding / RNA-loading assay presenting a defined miRNA duplex to recombinant CSR-1
Structural basis of guide 5'-ppp vs 5'-p selectivity in CSR-1	Family architecture (PAZ+PIWI) known; MID-pocket specificity not experimentally mapped for CSR-1	Would explain <i>why</i> CSR-1 favors 22G-RNAs and whether it could sterically accommodate a miRNA	Cryo-EM/crystal structure or MID-domain mutagenesis + loading assay
Whether any CSR-1 IP has ever recovered miRNAs above background	Published IPs report 22G-RNAs; miRNA fraction not explicitly quantified/excluded	A clean negative (no miRNA enrichment) would strengthen removal	Re-analysis of existing CSR-1 IP small-RNA-seq for miRNA reads vs input
Provenance completeness of the IBA term — RESOLVED	QuickGO withFrom = PTN001113179 confirmed; anchors alg-1/alg-2 + bilaterian AGOs	Confirms carry-over; ensures no hidden experimental support	Already resolved by QuickGO audit; leaf PTN000527417 not separately dereferenced

Method transparency: findings rest on public UniProt, PANTHER v19, and QuickGO records plus the cited PubMed abstracts. Where a value came from a database record (evidence codes, withFrom anchors) versus a primary experiment (guide-class identity, slicer activity), that distinction is preserved throughout. No in-house sequence/structure computation was fabricated; PANTHER internal-node annotation was retrieved via public API.

Discriminating Tests

The single most decisive experiment would be a **direct guide-loading / binding assay** with recombinant CSR-1A (and CSR-1B) presented with (a) a defined 22G-RNA (5'-ppp), (b) a canonical worm miRNA (e.g. let-7, 5'-p, Dicer-processed), and (c) a scrambled control, read out by EMSA/filter binding and by small-RNA-seq of what stays bound. Predicted outcome under the refutation: robust 22G-RNA binding, little-to-no specific miRNA binding.

Complementary discriminators:

1. **Re-mine existing CSR-1 IP small-RNA-seq** (Claycomb 2009; Nguyen & Phillips 2021 datasets) for canonical miRNA reads (5'-monophosphate, mature miRBase sequences) relative to input — a quantitative negative would convert "unsupported" toward "excluded."
 2. **5'-end chemistry dependence:** test whether a 5'-triphosphate is required for stable CSR-1 loading (predicts miRNA exclusion).
 3. **MID-pocket swap / point mutants** between CSR-1 and ALG-1 to test whether guide 5'-phosphate discrimination is the barrier to miRNA loading.
 4. **Comparative annotation audit:** map which nodes in PTHR22891 carry IDA miRNA-binding evidence versus IBA-only leaves to quantify propagation.
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Proposed Follow-up Actions / Curation Leads (require curator verification)

Lead 1 — Down-weight GO:0035198 "miRNA binding." Action: change from a supported/ asserted MF to **non-core**, or apply a **NOT** qualifier, or remove. Rationale: IBA-only, propagated from PTN001113179 whose anchors are alg-1/alg-2 and bilaterian AGOs; no worm experiment for CSR-1. Candidate reference to cite for removal rationale: QuickGO annotation record for H2KZD5/GO:0035198 (evidence ECO:0000318, GO_REF:0000033, withFrom PANTHER:PTN001113179).

Lead 2 — Prefer an informative siRNA-binding term. Add/prefer **GO:0035197 "siRNA binding" (22G-RNA binding)** over "miRNA binding" as the informative MF binding term; avoid falling back to generic "protein binding." Snippet to verify — [PMID: 19804758](#): "the CSR-1-

interacting small RNAs (22G-RNAs) are antisense to thousands of germline-expressed protein-coding genes." Snippet to verify — [PMID: 34244496](#): "*CSR-1 is an essential Argonaute protein that binds to a subclass of 22G-RNAs targeting most germline-expressed genes.*"

Lead 3 — Retain the experimentally grounded slicer function. Keep [GO:0070551](#) (endoribonuclease activity, cleaving siRNA-paired mRNA; IDA). Snippet to verify — [PMID: 18007599](#): "*An Argonaute protein, CSR-1, is responsible for the Slicer activity induced by secondary-type siRNAs.*"

Lead 4 — Note function is transcript "tuning," not silencing. Snippet to verify — [PMID: 27020753](#): "*employing catalytic slicing activity to broadly tune, instead of silence, germline gene expression.*"

Lead 5 — Identity hygiene. Confirm the review is anchored on H2KZD5/Q27GU1 only; explicitly exclude Q17370 (*nhr-47*) and Q21992 (deleted, LARP-1) as sources of any *csr-1* annotation.

Suggested curator questions: - Is there *any* primary assay demonstrating CSR-1-miRNA binding, or is GO:0035198 solely IBA? (Our search: solely IBA.) - Is there value in an explicit NOT-qualifier to prevent re-propagation, versus silent removal?

Suggested experiment: direct CSR-1 miRNA-vs-22G-RNA loading assay on recombinant CSR-1A/CSR-1B (see Discriminating Tests).

Bottom Line for the Curator

The hypothesis "CSR-1 binds miRNAs" should **not** be adopted as a curated CSR-1 function. The GO:0035198 annotation is an evolutionary carry-over from the Argonaute family node PTN001113179, whose experimental support lies with ALG-1/ALG-2 and bilaterian AGOs — not CSR-1. All CSR-1 primary data describe 22G-RNA binding and siRNA-directed slicing. Recommend treating GO:0035198 as non-core/removable (optionally NOT-qualified), promoting an informative **siRNA/22G-RNA binding** term, and retaining the IDA-supported slicer activity (GO:0070551). The one honest caveat — no direct assay has physically excluded miRNA binding — argues for cautious wording ("unsupported"), not for retaining an unsupported positive annotation. Identity claims (Q17370 = *nhr-47*; Q21992 = deleted LARP-1) are confirmed and should guard against accession carry-over.

Provenance: UniProt REST queries for H2KZD5, Q27GU1, Q17370, Q21992; PANTHER v19 geneinfo (PTHR22891:SF181); QuickGO annotation API for H2KZD5/GO:0035198; and PubMed abstracts for the PMIDs cited above. Computational results reported conservatively; no result fabricated.

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